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ORIGINAL RESEARCH

Theoretical study on coordination of methanol clusters to 3-methyl-4-pyrimidone

Mayaliwa Muzomwe[1] · **Jules Tshishimbi Muya**[1,2] · **Chung Hoeil**[2] · **Okuma Emile Kasende**[1]

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Abstract Density functional theory (DFT), MP2, and couple cluster ab initio methods were employed to investigate the microsolvation of 3-methyl-4-pyrimidone (3M4P) surrounded by methanol (MeOH) molecules. Structures are analyzed based on hydrogen bonds with a focus on relative energies, interaction energies, hydrogen bond cooperativity, hydrogen bonding geometries, and redshifts in the frequencies of O–H and C=O stretching modes. Our results show that there is no preferential orientation of MeOH attacks on the carbonyle site of 3M4P; both *trans* and *cis* 3M4P-MeOH complexes have same chance to be observed. *cis* 3M4P-MeOH and 3M4P-MeOH complex in which MeOH is located on N lie 0.56 and 3.11 kJ/mol at CCSD(T)/6-31+G(d,p) (0.63 and 1.67 kJ/mol at MP2/6-311++G(d,p)) above *trans* 3M4P-MeOH. MeOH dimers form more stable 3M4P-$(MeOH)_2$ complexes compare to 3M4P-$(MeOH)_2$ complexes in which individual MeOH molecules bind to carbonyl and N. Relative energies of 3M4P-$(MeOH)_3$ computed using various DFT methods point out the complex formed by linear MeOH trimer along methyl group of 3M4P (*cis* 3M4P-$(MeOH)_3$) as lowest. Carbonyl group is predicted as preferential site for hydrogen bond interaction. Besides O–H…O and O–H…N hydrogen bonds, 3M4P-$(MeOH)_2$ and 3M4P-$(MeOH)_3$ complexes are also stabilized by H–O…H–C weak interactions.

Keywords 3-Methyl-4-pyrimidone · Methanol · H-bonded complexes

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✉ Mayaliwa Muzomwe
muzmay@yahoo.fr

✉ Jules Tshishimbi Muya
julescmuya.tmuya@gmail.com

✉ Okuma Emile Kasende
okumakasende@unikin.ac.cd

Chung Hoeil
hoeil@hanyang.ac.kr

[1] Faculty of Sciences, University of Kinshasa, B.P.190 Kinshasa XI, Democratic Republic of the Congo

[2] Department of Chemistry, College of Natural Sciences, Hanyang University, Seoul, South Korea

Introduction

The 3-methyl-4-pyrimidone (3M4P) molecule is attracting growing attention by numerous research groups by its properties as pyrimidine bases which are constituents of nucleic acid bases. The latter are heterocyclic polyfunctional bases carrying carbonyl groups and nitrogen atoms as possible hydrogen bond (HB) sites. Much attention has been devoted to the properties of HBs of nucleic acid bases because of their involvement in the processes of reproduction and transmission of genetic information in nature [1, 2]. Because of the ubiquitous role of noncovalent interactions in many biological processes [3–8], the stability of the H-bonding involving the carbonyl or nitrogen sites of nucleic acid bases may be examined in a pyrimidine base such as 3M4P carrying a carbonyl group and two nitrogen atoms as possible nucleophilic sites in which one of the nitrogen sites is methylated, thus reducing the possibilities of interaction.

Theoretical and experimental investigations have been performed in the past in an attempt to determine or predict the preferred interaction site of 3M4P as well as 1-methyl-2-pyrimidone (1M2P) with proton donors of different acidic strength [9–14]. The outcome of this investigation revealed

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that strong proton donor acids interact preferentially with the non-methylated nitrogen site, whereas weak proton donor acids attack on oxygen. A density functional theory (DFT) study performed to determine the stability of the complexes formed between either the N or O site of 3M4P and 1M2P molecules with Lewis acids has revealed a strong region-selectivity with borane and its derivatives preferring the nitrogen site in both pyrimidine isomers, while a preference for oxygen is observed for the alkali acids in the 3M4P molecule [14]. The complexation of 1M2P with these hard alkali acids does not show any discrimination between the two sites due to the presence of a continuous delocalized density region between the nitrogen and the oxygen atoms as illustrated by the 1M2P molecular electrostatic potential (MEP).

From the perspective to discriminate between O–H…N and O–H…O=C complexes of 3M4P, B3LYP/6-31+G(d) and matrix–isolation FT-IR spectroscopy investigation was performed for complexes of 3M4P and methanol (MeOH) [13]. Attention has been paid to vibrational modes of 3M4P sensitive to H bonds. A substantial frequency decrease of the ν(C=O) mode of 22 cm^{-1} contrary to that of the ν(C2=N1) mode of 5 cm^{-1}, as well as small frequency increases of the δ(C=O) mode of 8 cm^{-1} and the γ(C=O) mode of 7 cm^{-1}, suggests that MeOH is H-bonded to 3M4P preferentially by the carbonyl group. In addition, the binding energies and geometrical parameters computed at the B3LYP/6-31+G(d) level of theory have argued the preference of MeOH to interact with 3M4P at the carbonyl site than at the N1 atom of the ring.

In the same way, MEP and Fukui function of 3M4P previously reported [14] have shown a preferential HB interaction on carbonyl in agreement with energetic, spectroscopic, and geometrical parameters. However, the uncertainty remains from these maps to predict whether the MeOH would prefer to attack the C=O on the left or right side. Since the oxygen atom of the carbonyl group of 3M4P has two electronic pairs oriented in the "anti" and "syn" directions. An HB formation with the carbonyl oxygen atom of the 3M4P can take place in both the "anti" Ha and "syn" Hb configurations, defined by the two electronic pairs of this oxygen atom. In other words, although H-bonding between MeOH and 3M4P occurs preferentially at the carbonyl group, the most stable configuration still remains unknown.

Furthermore, it is worth noting that the interaction of 3MeOH in low concentration is expected to lead to the formation of essentially 1:1 stoichiometry complexes. The experimental determination of rate of complexation for two equilibria in solution, and the experimental estimation of rate of complexation for more than two equilibria and their relative structural geometries is an arduous and more difficult task. The occurrences of various associations, whose proportion vary directly with the composition of the mixture, and indirectly with the properties of the medium, make the problem difficult to be solved. When several complexes occur in the system, the analysis is generally more laborious. Thus, computational methods appear as a complementary way of determining complex structures of higher stoichiometry. To this point, to the best of our knowledge, and despite the biological importance of this molecule, neither experimental nor theoretical work has examined the 3M4P complex structures of higher stoichiometry. The present study is designed to investigate the geometrical structures of various complexes formed between 3M4P and one to three MeOH molecules. The 3M4P-$(MeOH)_{n\,=\,1–3}$ clusters are studied in order to elucidate the microsolvation of 3M4P with MeOH on basis of predicted relative stabilities, interaction energies, cooperativity effects, natural bond orbital (NBO), hydrogen bond geometries, and O–H and C=O vibrational stretching redshifts. The HB cooperativity discussed with the NBO data is expected to play a major role in the stabilization and the structures of these complexes.

Computational method

B3LYP-D3, M06-2X, B97D3, MN12SX, wB97XD, and MP2 methods with 6-311++G(d,p) were employed for geometries optimization and vibrational analysis of different 3M4P-$(MeOH)_n$ complexes with n varying from 1 to 3. The vibrational modes were characterized by real frequencies meaning that the minima were reached. All the electronic energies reported in the present paper were corrected by zero point vibrational energies (ZPE) and basis set superposition error (BSSE) using counterpoise correction method. To gauge the performance of DFT and MP2 methods, CCSD(T)/6-31+G(d,p) single-point calculations were carried out from CCSD/6-31+G(d,p)-optimized geometries.

The relative energies were computed as total energy differences of 3M4P-$(MeOH)_n$ complexes from optimized geometries, while the interaction energies were computed as energy difference of the 3M4P-$(MeOH)_n$ complex and the sum of the 3M4P and MeOH adduct, as relaxed within the geometry of 3M4P-$(MeOH)_n$. The main contributions of the interaction energies were obtained by means of SAPT2 program [15] at HF/6-31+G(d,p) level as implemented in Q-Chem [16]. Natural bond orbitals [17] were analyzed to carry out the stabilization energy gains by delocalization of electron from the lone pair localized bonding orbital at the heteroatom of 3M4P to the antibonding orbital involving H–O bond of MeOH. The cooperative energy (CE) in 3M4P-$(MeOH)_n$ was defined as the difference between the total interaction energy of 3M4P-$(MeOH)_n$ and the sum of the interaction energies of 3M4P…(MeOH) and MeOH…MeOH pairwise interactions. When the cooperative energy was found negative (positive), the hydrogen bond formation was considered as cooperation (anti-cooperation) in nature. Calculations were performed using Gaussian 09 D01 program [18], Gaussview 5 [19], and Chemcraft 1.7 [20], which were used for visualization.

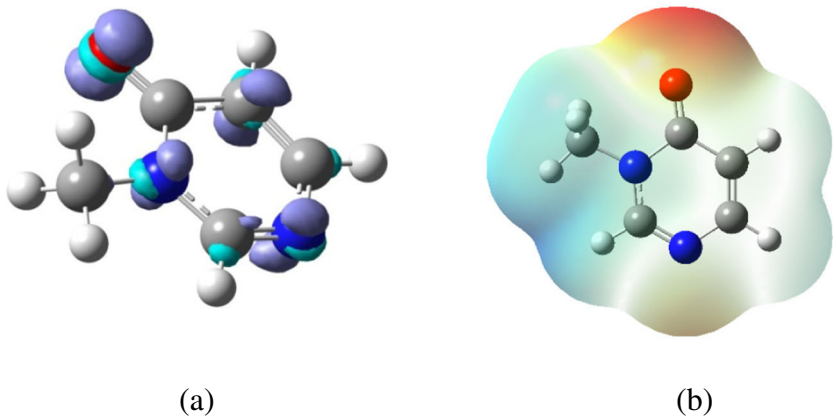

Fig. 1 Fukui function $f^-(\mathbf{r})$ (**a**) and MEP (**b**) of 3M4P (*gray* represents carbon (C); *white*, hydrogen (H); *blue*, nitrogen (N); and *red*, oxygen (O))

Results and discussion

Relative energies

Within conceptual DFT framework, Fukui function is referred as the derivative of the electron density of a molecular system with respect to its number of electrons at constant external potential. It describes how the electron density at different sites of the molecular system change when the total number of electrons of the molecular system change and shows the ability of atomic sites to gain and loose electrons upon nucleophilic and electrophilic attacks [21, 22]. The molecular electrostatic potential [22] is an alternative reactivity descriptor that provides information about where electrophilic and nucleophilic attacks are most likely to occur on a molecule. It results from the force acting on a proton located at given

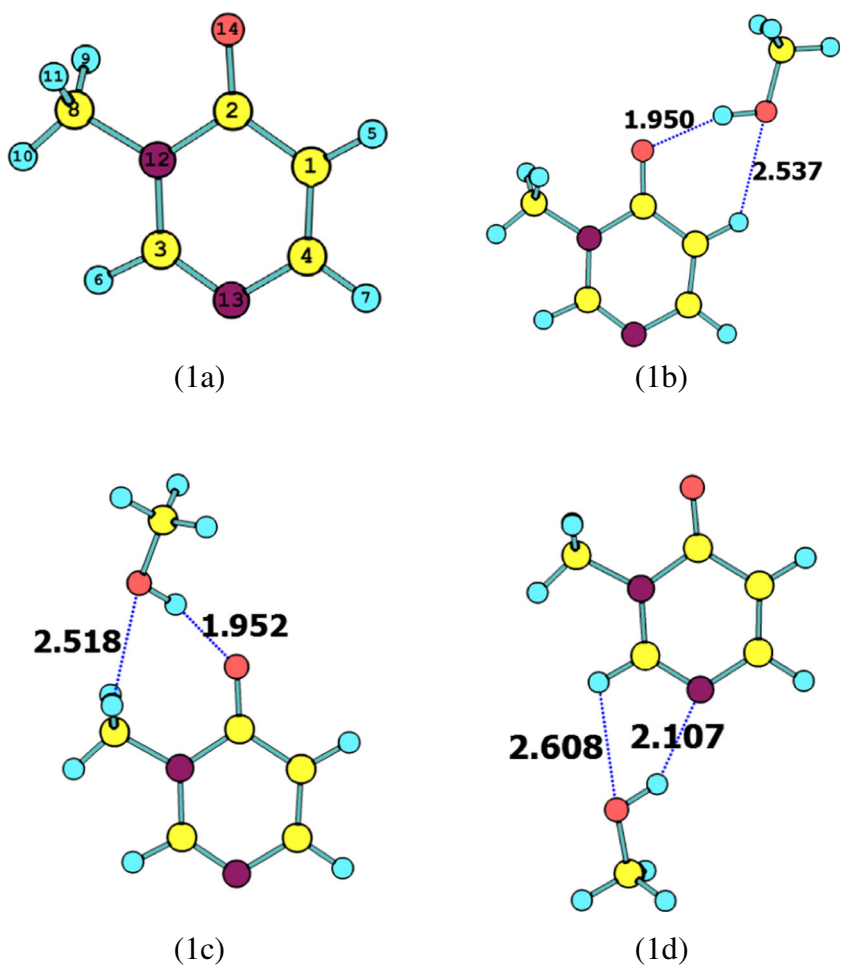

Fig. 2 3M4P(1a) and 3M4P-MeOH optimized (1**b**–**1d**) at CCSD/6-31+G(d,p)

distance from the molecular surface through the electron charge distribution generated by electrons and nuclei. The molecular electrostatic potential (MEP) and Fukui function of the 3M4P are provided in Fig. 1 in attempts to support the regioselectivity of this molecule. The MEP is widely used as good guide in predicting reactivity towards positively or negatively charged reactants. The most susceptible site for an electrophilic attack is the carbonyl where $f^{-}(\mathbf{r})$ is large and MEP has a negative large value. However, doubt still remains from these maps about the preferential direction of MeOH attack on C=O. All atoms of 3M4P are numbered for comparison purpose (**1a**). Figure 2 presents the hydrogen bond lengths of CCSD/6-31+G(d,p) 3M4P-MeOH-optimized geometries. **1b** and **1c** geometries differ by the orientation of MeOH with respect to O14 and are referred as *trans* and *cis*. These two geometries are stabilized by O_{14}…H–O, O…H_5–C_1, and O…H_9–C_8 hydrogen bonds. In **1d**, MeOH forms with 3M4P N_{13}…H–O and O...H_6–C_3 hydrogen bonds. Table 1 presents the relative stabilities of 3M4P-$(MeOH)_n$ complexes calculated at different theory levels. All the data listed in this table are BSSE and ZPE corrected, except CCSD and CCSD(T) values that include only BSSE correction. Two competitive low-laying geometries (**1b** and **1c)** were identified that differ by 0.56 kJ/mol at CCSD(T)/6-31+G(d,p) (0.64 kJ/mol at MP2/6-311++G(d,p)). This energy difference is small and clearly indicates there is no preferential orientation of MeOH around O14; both *cis* and *trans* complexes (**1b** and **1c**) are assumed to have same chance to be observed. **1d** lies 3.11 kJ/mol at CCSD(T)/6-31+G(d,p) above **1c** (1.67 kJ/mol at MP2/6-311++G(d,p)). The relative energies of **1b** and **1d** computed at B3LYP-D3 and MP2 are in good agreement with CCSD(T). We were intrigued to see where additional methanol molecule in 3M4P-MeOH would prefer to bind. For this purpose, 3M4P-$(MeOH)_2$ and 3M4P-$(MeOH)_3$ complexes were studied. Figures 3 and 4 show the equilibrium geometries of 3M4P-$(MeOH)_2$ and 3M4P-$(MeOH)_3$ complexes. The relative energies of 3M4P-$(MeOH)_2$ complexes with respect to the ground state computed at MP2/6-311++G(d,p) as summarized in Table 1 decrease in the following order: **2d** ~ **2e** < **2c** < **2b** < **2a**. Like **1b** and **1c,** the energy difference between the two lowest geometries (**2d** and **2e**) is about 0.03 kJ/mol at MP2/6-311++G(d,p) (1.28 kJ/mol at B3LYP-D3, lies within thermal energy) and may suggest the two complexes to have same relative stability. The relative energies of 3M4P-$(MeOH)_3$ computed using various DFT methods point out **3e** as the lowest. MeOH molecules in 3M4P-$(MeOH)_3$ prefer to interact with carbonyle of 3M4P in such a way they can bind to each other through O–H..O hydrogen bonds to form a linear methanol trimer. The energy difference between **3e** and **3f** lies in between 8.18 and 4.63 kJ/mol. The difference in relative stabilities between **3a** and **3b** like that between **3c** and **3d** is quite small and underlines once again that the *cis* or *trans* orientation of MeOH clusters on O14 does not change significantly the total electronic energy of the 3M4P-$(MeOH)_n$ complexes.

Interaction and cooperativity energies

The interaction energies between 3M4P and MeOH are given in Table 2. The data in this table are self-explanatory. The interaction energies between MeOH and 3M4P in 1**b**, 1**c**, and 1**d** amount to be −28.85, −29.40, and −26.04 kJ/mol at CCSD(T)/6-31+G(d,p) (−28.01, −28.52, and −26.29 kJ/mol at MP2/6-311++G(d,p)). With respect to the CCSD(T) reference values, CCSD and MP2 perform quite well compare to DFT methods. The SAPT2 interaction energies for **1b**, **1c**, and **1d** estimated about −34.37, −36.45, and −32.07 kJ/mol are approaching M062X, B3LYP-D3, and wB97XD values. The

Table 1 Relative energies to the ground state in kJ/mol of 3M4P-$(MeOH)_n$ including BSSE and ZPE corrections computed using various DFT methods and MP2 at 6-311++G(d,p). CCSD and CCSD(T) were computed at 6-31+G(d,p)

	B3LYP-D3	wB97XD	B97D3	M062X	MNS12X	MP2	CCSD	CCSD(T)
1b	0.62	1.62	0.07	1.71	1.85	0.63	0.64	0.56
1c	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1d	2.89	3.59	0.24	4.66	4.89	1.67	3.06	3.11
2a	14.69	13.81	13.40	9.90	8.85	14.66		
2b	14.70	13.24	13.16	11.10	10.12	14.50		
2c	12.36	11.10	11.98	8.66	7.29	13.81		
2d	0.00	0.02	0.00	3.74	3.75	0.03		
2e	1.28	0.00	1.47	0.00	0.00	0.00		
3a	19.77	20.29	15.44	19.51	16.28			
3b	22.57	–	–	–	–			
3c	20.43	19.74	16.59	20.83	15.35			
3d	18.62	18.81	15.45	21.64	16.83			
3e	0.00	0.00	0.00	0.00	0.00			
3f	7.82	7.37	4.63	8.18	5.03			

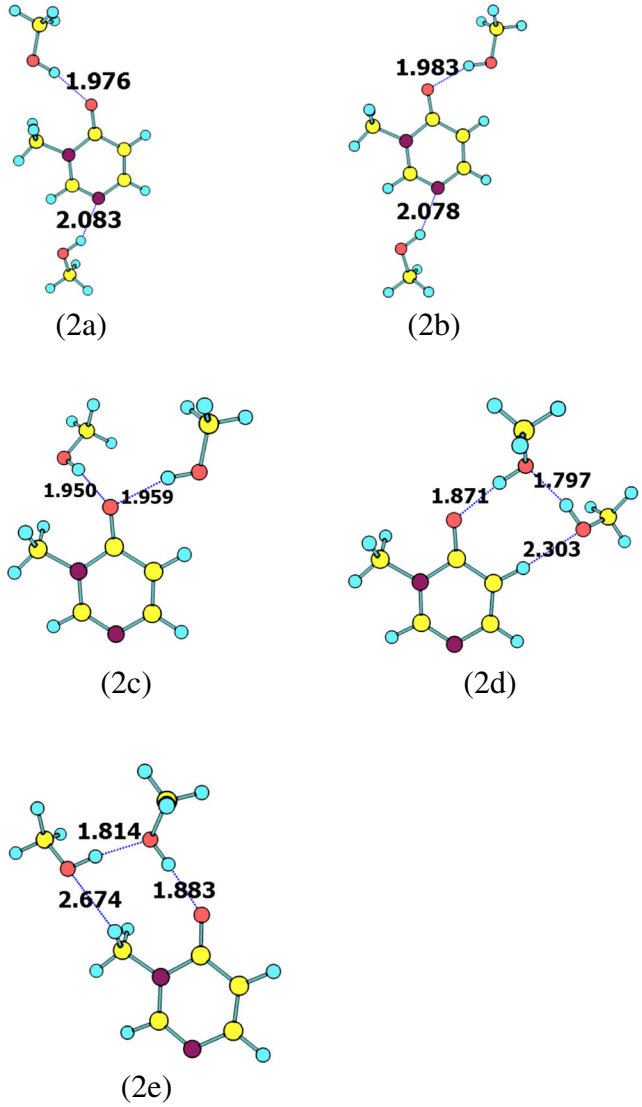

Fig. 3 Optimized geometries of 3M4P-$(MeOH)_2$ computed at MP2/6-311+G(d,p)

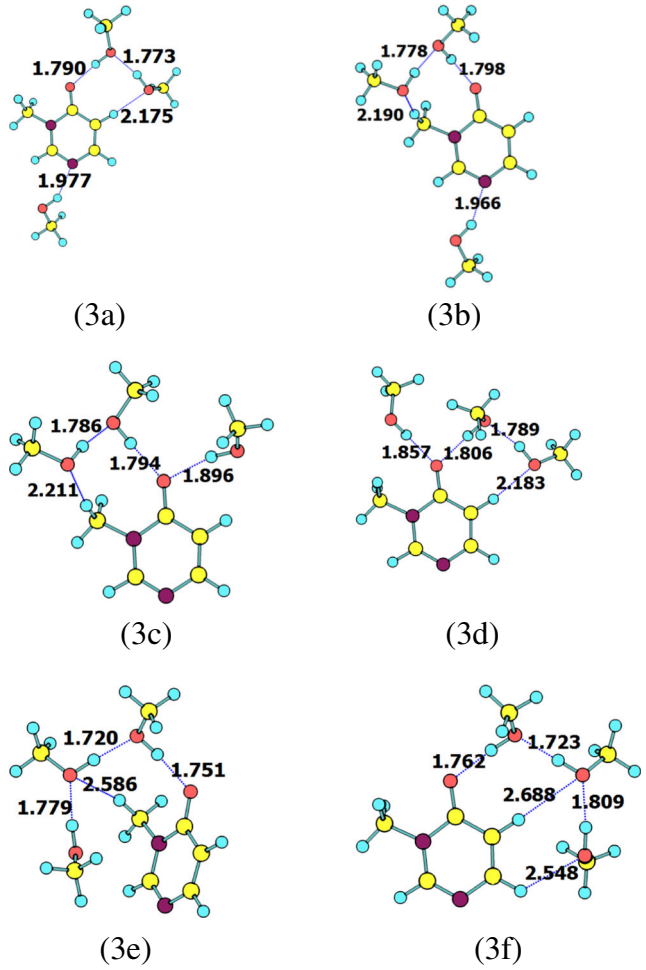

Fig. 4 Optimized geometries of 3M4P-$(MeOH)_3$ complexes computed at B3LYP-D3/6-311++G(d,p)

main contribution in SAPT2 (HF) interaction energies as shown in Table S2 arises from coulombic interactions which are about four times larger than the induced energies and the dispersion energies. This is understandable, because hydrogen bonding is fundamentally known as an electrostatic interaction. It is demonstrated that hydrogen bonds can act cooperatively either to reinforce or weaken the hydrogen bond framework. The cooperativity energies of 3M4P-$(MeOH)_2$ and 3M4P-$(MeOH)_3$ complexes computed at MP2 and B3LYP-D3 are collected in Table 3. **2d** and **2e** complexes show cooperativity energies of −2.16 and −2.21 kJ/mol and are considered to have hydrogen bonding acting cooperatively. The establishment of hydrogen bonds in **2e** and **2d** goes together with a relative cyclisation of MeOH molecules, and C–H…O hydrogen bond interactions might enhance the stability of those complexes. **2a**, **2b**, and **2c** are characterized by an anti-cooperativity effect of hydrogen bonding. However, simultaneous complexation on carbonyl by two MeOH molecules, as the case of **2c**, yields positive cooperativity energy. The cooperativity effect is more significant in **3e** and **3f** (~ −4 kJ/mol at B3LYP-D3 level) compare to **3a**, **3c**, and **3d** (~

Table 2 Interaction energies in parenthesis in kJ/mol of 3M4P-MeOH including BSSE and ZPE corrections computed using various DFT methods and MP2 at 6-311++G(d,p). CCSD, CCSD(T), and SAPT2(HF) were computed at 6-31+G(d,p)

	1b	**1c**	**1d**
B3LYP-D3	−35.91	−36.71	−32.97
wB97XD	−34.27	−35.66	−31.58
M062X	−34.73	−36.79	−32.38
B97D3	−31.71	−32.09	−30.79
MNS12X	−31.46	−33.56	−27.72
MP2	−28.01	−28.52	−26.29
CCSD	−27.89	−28.48	−25.16
CCSD(T)	−28.85	−29.40	−26.04
SAPT2	−34.30	−36.45	−32.07

Table 3 Cooperativity energies (CE in kJ/mol) and pairwise energies (E*ij*) between fragments *i* and *j* of 3M4P-$(MeOH)_2$ and 3M4P-$(MeOH)_3$ at MP2 and B3LYP-D3 at 6-311++G(d,p)

Complexes	E	E12	E13	E14	E23	E24	E34	CE
2a	−16.2	−8.74	−7.68	–	0.14	–	–	0.08
2b	−15.92	−8.22	−7.66	–	−0.12	–	–	0.08
2c	−16.65	−8.95	−8.45	–	−0.64	–	–	1.39
2d	−17.87	−7.64	−2.58	–	−5.49	–	–	−2.16
2e	−18.82	−2.79	−8.41	–	−5.41	–	–	−2.21
3a	−26.65	−5.79	−7.7	−5.98	0.15	−5.66	−0.04	−1.63
3b	–	–	–	–	–	–	–	–
3c	−26.63	−8.12	−3.03	−8.35	−5.52	0.54	0.06	−2.21
3d	−26.24	−8.94	−2.68	−7.72	0.15	0.47	−5.7	−1.82
3e	−31.21	−1.98	−4.32	−7.7	−5.57	−5.67	−1.8	−4.17
3f	−28.88	−2.15	−2.46	−7.79	−5.82	−5.42	−0.96	−4.28

E is the binding energies of 3M4P-$(MeOH)_2$ or 3M4P-$(MeOH)_3$

i = 1 describes 3M4P fragment

i = {2,3,4} represents MeOH fragment

−2 kJ/mol). The cooperativity energies trend is in good agreement with the computed relative stability of complexes.

NBO second-order perturbation energies, E(2)

The quantities of charge transferred from lone pair donor orbitals (O(LPs) and N(LP)) to the antibonding acceptor orbitals (H–O and C–H bonds) in 3M4P-MeOH and 3M4P-$(MeOH)_2$ complexes were estimated using NBO perturbation theory E(2) at MP2 and B3LYP-D3 with 6-311++G(d,p) (Table 4). Figure S1 provides labels used to number all various atoms in 3M4P-MeOH and 3M4P-$(MeOH)_2$ complexes. The stabilization energy between the donor O14(LPs) and the acceptor H–O(σ*) in **1b** assessed about 30.29 kJ/mol gives the strongest interaction compare to O14(LPs)→(σ*)H–O in **1c** (E(2) of 26.94 kJ/mol) and N(LP)→(σ*)H–O interaction in **1d** (E(2) of 26.15 kJ/mol). Nonetheless, the NBO E(2) energy values associated with O14(LPs)→(σ*)H–O in **1b** and **1c** are very close and support the idea of competition of **1b** and **1c** geometries in low-lying energy states already mentioned above. All 3M4P-MeOH complexes studied in the present work benefit from a slight increase in stability from secondary O(LPs)→C–H(σ*) weak interactions (E(2) < 4 kJ/mol) between methanol and 3M4P. The C1–H5…O interaction in **1b** gives the strongest C–H...O stabilization energy of 2.89 kJ/mol (Table 4). The second-order perturbation theory analysis computed in NBO basis at B3LYP-D3 level shows a significant amount of E(2) charge transfer energy in **2d** and **2e** estimated about 61.09 and 37.36 kJ/mol arising from O14(LPs) to O–H σ*-bond. The O(LPs)→(σ*)H–O interactions between MeOHs in those complexes yield stabilization E(2) energies of 61.96 and 60.46 kJ/mol. This strong O–H…O hydrogen bond between MeOHs is not present in **2a**, **2b**, and **2c**. The E(2) charge transfer in **2c** involves O14(LPs) and O–H (σ*-bond) and amounts to be 61.00 kJ/mol. The N(LP)→(σ*)H–O interactions in **2a** and **2b** are characterized by E(2) energies of 22.59 and 33.81 kJ/mol. The O–H…O and C–H…O hydrogen bonds formed between MeOH and 3M4P combined with the O–H…O interactions in between MeOHs are responsible of the high stability of **2d** and **2e**.

Table 4 NBO second-order perturbation energy E(2) (kJ/mol) of the hydrogen bonding between 3M4P and MeOH in 3M4P-MeOH computed at MP2/6-311++G(d,p) and in 3M4P-$(MeOH)_2$ computed at B3LYP-D3/6-311++G(d,p). Labels of atoms are displayed in Fig. S1

Entry	Charge transfers	NBO E(2)
1b	O14(LP)-> H20-O15(σ*)	30.29
	O15(LP)-> C1-H5(σ*)	2.89
1c	O14(LP)-> H20-O15(σ*)	26.94
	O15(LP)-> C8-H9(σ*)	1.09
1d	N13(LP1)-> H20-O15(σ*)	26.15
	O15(LP1)-> C3-H6(σ*)	1.00
2a	O14(LP)-> H20-O15(σ*)	16.69
	N13(LP)-> O15-H20(σ*)	22.59
2b	O14(LP)-> H20-O15(σ*)	39.08
	N13(LP)-> H20-O15(σ*)	33.81
2c	O14(LP)-> H20-O15(σ*)	23.01
	O14(LP)-> H20-O15(σ*)	37.99
2d	O14(LP)-> H20-O15(σ*)	61.09
	O14(LP)-> C1-H5(σ*)	17.15
	O15(LP)-> H21-O22(σ*)	61.97
2e	O14(LP)-> H20-O15(σ*)	37.36
	O22(LP)-> C8-H9(σ*)	14.60
	O15(LP)-> H21-O22(σ*)	60.46

Geometrical and spectroscopic properties (redshift)

The C–H…O bonds in **1b**–**d** geometries as shown in Fig. 2 are longer than 2.5 Å and are assumed to be less strong. The O–H…O (1.95 Å) in **1b**–**c** and O–H…N (2.1 Å) in **1d** are quite

short and contribute quite significantly to the stabilization of 3M4P-MeOH complexes than C–H…O bonds. According to the geometric data listed in Table S1, the O–H bonds of MeOH involved in hydrogen bonding interactions are slightly elongated (MeOH has a O–H bond of 0.962 Å at CCSD/6-31+G(d,p)). But the overall picture of the geometrical features of hydrogen bond distances in 3M4P-MeOH are quite cumbersome, difficult to predict which complex will be the most stable based on O–H, O..H, and CH…O bonds and OH...O angles. The elongation of O–H bond of MeOH involved in the hydrogen bond was also observed by Muzomwe et al. [12, 13]. The MP2/6-311+G(d,p)-optimized geometries of 3M4P-$(MeOH)_2$ are depicted in Fig. 3. In this figure, **2d** and **2e** have the shortest hydrogen bond lengths. Based on the hydrogen bond distances, one can predict the following increasing rank of stability: **2a** ~ **2b** < **2c** < **2e** < **2d**. This order of stability is in agreement with the computed rank of relative energies discussed above. The O–H…O bond distances formed by MeOH–MeOH interactions are shorter compare to the O–H…O14 bond distances resulting from the interaction between 3M4P and MeOH. Figure 4 shows the hydrogen bond lengths of six 3M4P complexes comprising three explicit MeOH molecules. **3c** and **3d** equilibrium geometries have each a methanol dimer connected by an intermolecular O…H–O bond of 1.79 Å and are respectively bound to 3M4P through O14…O–H bond of 1.79 and 1.81 Å, and O…H–C bond of 2.21 and 2.18 Å. **3a** and **3b** are analogous to **3d** and **3c**; both structures have methanol dimers stabilized by O…H–O and O…H–C bonds but differ by the position of the third MeOH, which is linked to O14 by O14…H–O (1.86 and 1.90 Å) in **3d** and **3c**, whereas in **3a** and **3b**, it is attached to N3 through N3…H–O (1.98 and 1.97 Å). It is possible that this third MeOH does not bind to neither O14 nor N3 but prefer to interact with a methanol dimer to form a linear methanol trimer (**3e** and **3f**). The latter are the lowest energy geometries and comprise the shortest O–H…O bond between MeOHs of 1.72 Å.

The hydrogen bond strength can also be probed by measuring the vibrational shift of O–H and C=O stretching modes. Density functional theory is known to provide more reliable vibrational spectra of hydrogen bonding complexes. The spectroscopic data of 3M4P-$(MeOH)_{n\ =\ 1-3}$ complexes calculated at B3LYP/6-31++G(d,p) are given in Table S3. The redshifted frequencies of O–H in 3M4P-MeOH suggest the MeOH to interact stronger on $C{=}O_{14}$ (190, 193 cm^{-1}) than N13 (141 cm^{-1}). The bathochromic effect of νC=O vibrators is actually more important in complex where MeOH molecules form hydrogen bonds with the two carbonyl electron pairs (**2c**, **3c**, and **3d**). The analysis of the vibrational shifts shows large redshifted in O–H frequency for complexes formed by methanol dimer and trimer interacting with 3M4P such as **2d**, **2e**, **3e**, and **3f**. The magnitude of the computed red frequency shifts ($\Delta\nu_{OH}$ and νC=O) in complexes is in good agreement with the relative stabilities of complexes discussed in previous sections.

Conclusion

3M4P surrounded by one to three MeOH molecules has been investigated by ab initio and DFT methods with 6-31+G(d,p) and 6-311++G(d,p) basis sets. Our results show that **1b** and **1c** (**2d** and **2e**) are lowest with competitive relative stabilities. Unlike 3M4P-MeOH and 3M4P$(MeOH)_2$ complexes which have no preference in the orientation of MeOH on the carbonyle, in 3M4P-$(MeOH)_3$ complexes, the *cis* form becomes more favored. The relative energies of 3M4P-$(MeOH)_3$ computed using various DFT methods point out **3e** as the lowest geometry. MeOH reacts preferentially on carbonyl site of 3M4P forming a linear chain of methanol adducts stabilized by O–H…O and C–H…O hydrogen bonds. The relative energies, interaction energies, cooperativity energies, intermolecular distances, and frequency shifts are used in the discrimination of 3M4P-$(MeOH)_{n\ =\ \{1,2,3\}}$.

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Conflict of interest The authors declare that they have no conflict of interest.

References

1. Alberts B, Johnson A, Lewis J, et al. (2002) Molecular biology of the cell, 4th edn. Garland Science, New York,
2. Lodish H, Berk A, Zipursky SL, et al. (2000) Molecular cell biology, 4th edn. W. H. Freeman, New York,
3. Jeffrey GA, Saenger W (1991) Hydrogen bonding in biological structures. Springer-Verlag, Berlin,
4. Jeffrey GA (1997) An introduction to hydrogen bonding. Oxford University Press, New York,
5. Grabowski SJ (ed.) (2006) Hydrogen bonding—new insights. Springer, New York,
6. Scheiner S (1997) Hydrogen bonding, a theoretical perspective. Oxford University Press, New York,
7. Szatyłowicz H, Sadlej-Sosnowska N (2010) J Chem Inf Model 50: 2151–2161
8. Matta CF, Castillo N, Boyd RJ (2006) Extended weak bonding interactions in DNA: π-stacking (base-base), base-backbone, and backbone-backbone interactions J Phys Chem B 110:563–578
9. Kasende O, Zeegers-Huyskens T (1981) J Mol Struct 75:201–207
10. Kasende O, Zeegers-Huyskens T (1984) J Phys Chem 88:2636–2642
11. Kasende O, Zeegers-Huyskens T (1984) Spectrosc Lett 17:783–801
12. Muzomwe M, Boeckx B, Maes G, Kasende OE (2011) S Afr J Chem 64:23–33
13. Muzomwe M, Maes G, Kasende OE (2012) Nat Sci 4:286–297

14. Kasende OE, Muya JT, Broeckaert L, Maes G, Geerlings P (2011) J Phys Chem A 116:8608–8614
15. Szalewicz K (2012) Wiley interdisciplinary reviews: computational molecular science, vol 2, pp. 254–272
16. Shao Y, Gan Z, Epifanovsky E, Gilbert ATB, et al. (2015) Mol Phys 113:184–215
17. Weinhold F, Landis CR (2012) Discovering chemistry with natural bond orbitals. John Wiley & Sons, Hoboken, New Jersey,
18. Gaussian 09, Revision D.01, M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery, Jr., J. *E. Peralta*, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V. N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J. M. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, Ö. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski, and D. J. Fox, Gaussian, Inc., Wallingford CT, (2009).
19. GaussView, Version 5, Dennington, R.; Keith, T.; Millam, J., *Semichem Inc.*, Shawnee Mission, KS (2009).
20. http://www.chemcraftprog.com
21. Geerlings P, Proft FD, Langenaeker W (2003) Conceptual density functional theory Chem Rev
22. Murray JS, Politzer P (2011) The electrostatic potential : an overview Commput Mol Sci 1:153–163